Machine-Checked Algorithmic Consequences of a $2^{o(n)}$ Algorithm for #k-SAT

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Abstract

In October 2026 two groups claimed algorithms for #k-SAT in time $2^{o(n)}$, which would refute the Exponential Time Hypothesis, and both formalized their algorithms in Lean. A refuted lower bound yields no algorithm by itself: every consequence needs a reduction to #SAT whose number of variables is linear in the relevant parameter. We give such reductions for twenty-four problems and prove each resulting algorithm as a Lean 4 theorem about a single Turing machine, on top of Anthropic's formalization and with the standard axioms only. The new results are: the parity of the number of connected vertex sets of size ℓ that satisfy a given CNF, on graphs of any density; from it, with an isolation lemma for arbitrary sets, one-sided-error algorithms for Steiner tree and connected vertex cover on graphs of any density; the parity and a one-sided-error decision for connected dominating set on graphs of any density, whose clauses of domination have unbounded width, by running the clause-shortening tree of Anthropic's dominating-set algorithm inside the Cut & Count formula, and from it Max Leaf Spanning Tree and minimum connected dominating set; the same for any positive CNF of unbounded width with linearly many clauses, which covers Group Steiner tree, connected hitting set and connected total domination, and, on connected graphs, the minimum versions of Steiner tree, connected vertex cover and Group Steiner tree; the parity and a one-sided-error decision for feedback vertex set on graphs with $O(n)$ edges, through a Cut & Count formula for induced forests; #CNF-SAT of unbounded clause width with up to $n \cdot 2^{(\lg n)^{1/4}/(\lg\lg n)^2}$ clauses; counting k-colorings for all k with $\lg k \le (\lg n)^{1/6-\theta}$; and the Max-SAT spectrum. We also formalize direct corollaries (homomorphisms into a target in the input, the density of states of the Potts model and the Tutte polynomial on the hyperbolas $(x-1)(y-1)=q$, the permanent of $\{-1,0,1\}$ matrices in $2^{\varepsilon m}\operatorname{poly}(n)$) and known consequences (#CNF-SAT of any width with linearly many clauses, the 0/1 permanent, Vertex Cover and Nonblocker in $2^{\varepsilon k}\operatorname{poly}(n)$, and odd cycle transversal on graphs of any density). Our results on Hamiltonian cycles, the parity for $o(n\sqrt{\lg n}/\lg\lg n)$ edges and one-sided error for $O(n)$ edges, were obtained independently of Tulsyan's concurrent exact count, which supersedes them. All bounds hold for every $\varepsilon > 0$. We also prove a barrier: over the rationals, Cut & Count with q colors whose weights are shared by all vertices cannot separate the connected vertex sets of size $\ell > q$ from the disconnected ones, which explains why our connectivity results are parities or have one-sided error. We give reference implementations of all reductions, tested against brute force and against the Lean definitions.

1 Introduction

The Exponential Time Hypothesis (ETH) of Impagliazzo and Paturi [16] states that 3-SAT on n variables needs time $2^{\Omega(n)}$. Much of fine-grained complexity rests on it: tight lower bounds for coloring, Hamiltonian cycle and Max-Cut, the $2^{O(\sqrt{n})}$ barrier for planar problems, and most lower bounds for parameterized problems. In October 2026 two groups claimed that ETH is false.

- OpenAI [21] claims #k-SAT for every fixed k in time

$$(n+m)^{O_k(1)} \cdot 2^{O_k(n \lg\lg n/\sqrt{\lg n})},$$

deterministic and in polynomial space, by contracting Boolean tensor networks with s wires in time $2^{O(s/\sqrt{\log s})}$.

- Anthropic [1] claims #k-SAT with one machine for all k in time

$$2^{C \min(k^2 n \lg\lg n/\sqrt{\lg n},\, km/\sqrt{\lg m})}$$

for an absolute constant C, by sum-product computations on the clause graph.

Both papers are unrefereed, and both come with Lean formalizations [2]. If the claims hold, ETH, SETH, #ETH, Gap-ETH and NSETH fall. The OV conjecture, the CNF-SAT hypothesis for poly(n) clauses of unbounded width, the k-Clique conjecture and the Set Cover conjecture still stand. The algorithms are galactic: Anthropic reports no gain over 2^n below $n \approx 10^{10}$.

Corollaries are not automatic. A refuted lower bound gives no algorithm. To obtain a $2^{o(n)}$ algorithm for a problem X from a $2^{o(n)}$ algorithm for #SAT, one needs a reduction from X to #SAT with $o(n\sqrt{\lg n}/\lg\lg n)$ variables and, for a uniform bound, small clause width. Problems in size-constrained SNP reduce in this way [17]. Problems with global constraints (connectivity, orderings, large numbers) or dense pairwise interactions do not, at least not directly. This paper is about the second kind of problem: we find reductions that keep the number of variables linear in the parameter even though the natural encoding does not.

Contributions.

1. *Connectivity on graphs of any density* (Section 3). The parity of the number of connected vertex sets of size ℓ that satisfy a CNF (N2), and from it one-sided-error algorithms for Steiner tree and connected vertex cover (N2′), on graphs with any number of edges. The same for connected dominating sets (N7, N7′), and from it Max Leaf Spanning Tree and minimum connected dominating set (N7″). More generally, the same for the connected solutions of any positive CNF of unbounded width with linearly many clauses (N8, N8′): Group Steiner tree, connected hitting set and connected total domination, and, on connected graphs, the minimum versions (N8″). These are, to our knowledge, the first consequences of the claimed refutations for connectivity problems on dense graphs; the concurrent work on Hamiltonian cycles [25] needs average degree $2^{o(\sqrt{\lg n})}$. On graphs with $O(n)$ edges, the parity of the number of feedback vertex sets of each size and a one-sided-error decision (N9, N9′), by Cut & Count for induced forests.

2. *Counting beyond the bounded-width case* (Section 4): #CNF-SAT of unbounded width with superlinearly many clauses (A1′), k-colorings for k growing with n (A2) and the Max-SAT spectrum (A4), with direct corollaries for homomorphisms, the Potts model and the Tutte polynomial, and permanents.

3. *Twenty-two algorithms* (Table 1; N7″ and N8″ reuse machines), each one a polynomial-time reduction to one of three counters from Anthropic's development, followed by a simple post-step. They are written out as pseudocode in Sections 3–6.

4. *Machine-checked statements.* Each algorithm is a Lean 4 theorem stating that one Turing machine (Mathlib's multi-stack model TM2, finite alphabets) halts on a given input with a given output within a given number of steps. The proofs build on Anthropic's development, which proves its $2^{o(n)}$ algorithm as a theorem; the results are therefore Lean theorems, not conditional statements. They use only the axioms `propext`, `Classical.choice` and `Quot.sound`.

5. *A barrier for Cut & Count* (Section 7), formalized: with q color weights shared by all vertices, no rational linear combination of the counts separates the connected sets of size $\ell > q$ from the disconnected ones, and $q = \ell$ colors would cost $\lg \ell$ bits per vertex.

6. *The isolation lemma of [6] for arbitrary sets*, formalized: sparse parity checks on k coordinates isolate a point of any nonempty $S \subseteq \{0,1\}^M$ with probability $2^{-\delta M}$. The proof of [6] is oblivious to the formula [3, §9.3]; Tulsyan [25] formalizes the same lemma. Unlike Valiant–Vazirani hashing [26], it keeps the number of variables linear.

7. *Reference implementations* of all reductions in Python. They build the same formulas from the same input words as the Lean machines, and they are tested against brute force and against the Lean definitions.

Techniques. Three ideas recur. First, Anthropic's *sized* counter, which counts the solutions with exactly s true variables, can be combined with complement bits $q = \neg p$: every complementary pair contributes exactly one true variable, so auxiliary variables leave the target size fixed. This is how side bits, isolation checks and parity chains enter a sized count (N2, N2′, N6′, N9, N10). Second, Cut & Count [10] turns connectivity into the parity of a count, and with a sized counter the number of edges no longer matters. Third, search trees (clause shortening for A1 and, inside a Cut & Count formula, for N7 and N8; degree branching for A3) reduce the parameter or the width before counting at the leaves.

Honest labels. Table 1 marks each result, after literature searches of 28 and 29 October 2026 (papers, preprints, blogs and repositories). A1, A3, A3′, N6 and N10 are known consequences that we formalize. N3, N5 and N6′ are direct corollaries: the reduction is the obvious one. A5a, A5a′ and A5b are superseded by concurrent work [25]. The remaining results, N2, N2′, N7, N7′, N7″, N8, N8′, N8″, N9, N9′, A1′, the range of k in A2 and the spectrum in A4, are ours: we did not find them, which does not mean that they are certainly new.

Concurrent work. Tulsyan [25], independently and also on top of Anthropic's Lean development, proves that one machine computes the number of Hamiltonian cycles of a graph with n vertices and m edges exactly and deterministically within $C2^{C(m+n)/\sqrt{\lg(n+2)}}$ steps, and within $C2^{\delta n}$ steps for $m \le Dn$. His method uses Anthropic's network machinery together with a matrix-tree identity, rather than the CNF counter. An exact count yields the parity and a deterministic decision, so his result supersedes A5a, A5a′ and A5b. Our versions were posted some hours earlier, on 27 October; we state them briefly in Section 5. We rebuilt both theorems from his repository on the same revision of Anthropic's development; they compile and use only the standard axioms, and his count is the same quantity as our $\mathrm{hc}(G)$ of Section 5. Since then (29 October) he has added a Lean-verified exact algorithm for TSP with polynomial weights on graphs with $m \le Dn$, a Lean-verified exact count in $2^{O(n \log \bar{d}/\sqrt{\lg n})}$ for average degree $\bar{d}$, Lean proofs of decision and search for Hamiltonian cycles and of Hamiltonian paths, and Lean-verified exact directed Hamiltonian cycle and ATSP. He also sketches deterministic decision for Steiner tree, feedback vertex set, connected vertex cover,

Table 1: The results. Time bounds hold for every $\varepsilon > 0$ (resp. $\delta > 0$); C and n_0 depend only on the stated parameters. m is the number of edges, clauses or nonzero entries.

	Problem and bound	Status
N2	$\oplus$connected vertex sets of size ℓ satisfying a CNF, any density; $C2^{\delta n}$	ours
N2′	Steiner tree, connected vertex cover, one-sided error, any density; $C2^{\varepsilon n}$	ours
N7	$\oplus$connected dominating sets of size ℓ, any density; $C2^{\delta n}$	ours
N7′	connected dominating set, one-sided error, any density; $C2^{\varepsilon n}$	ours
N7″	Max Leaf Spanning Tree, minimum connected dominating set, one-sided error, any density; $C2^{\varepsilon n}$	ours
N8	$\oplus$connected vertex sets of size ℓ meeting $\le Dn$ given sets, any density; $C2^{\delta n}$	ours
N8′	Group Steiner tree, connected hitting set, connected total domination, one-sided error, any density; $C2^{\varepsilon n}$	ours
N8″	minimum Steiner tree, connected vertex cover, Group Steiner tree, one-sided error, connected G of any density; $C2^{\varepsilon n}$	ours
N9	$\oplus$feedback vertex sets of size k, $m \le Dn$; $C2^{\delta n}$	ours
N9′	feedback vertex set, one-sided error, $m \le Dn$; $C2^{\varepsilon n}$	ours
A1	#CNF-SAT, any width, $m \le Dn$; $C2^{\delta n}$	known [7, 1]
A1′	#CNF-SAT, any width, $m \le n\,2^{(\lg n)^{1/4}/(\lg\lg n)^2}$, one machine; $2^{\delta n}$	ours
A2	#k-colorings, all k with $\lg k \le (\lg n)^{1/6-\theta}$, one machine; $2^{\varepsilon n}$	ours (range of k)
N5	#homomorphisms $G \to H$, H in the input, $\lg\lvert V(H)\rvert \le (\lg n)^{1/6-\theta}$; $2^{\varepsilon n}$	corollary
A4	Max-SAT spectrum, width r, $m \le Dn$; $C2^{\delta n}$	ours (spectrum)
N3	Potts density of states, Tutte polynomial on $(x-1)(y-1) = q$, $m \le Dn$; $C2^{\delta n}$	corollary
N6	0/1 permanent; $C(n+2)^C 2^{\varepsilon m}$	known [21]
N6′	$\{-1,0,1\}$ permanent; $C(n+2)^C 2^{\varepsilon m}$	corollary
A5a	$\oplus$Hamiltonian cycles, degree $\le \Delta$; $2^{\varepsilon n}$	superseded [25]
A5a′	$\oplus$Hamiltonian cycles, $m = o(n\sqrt{\lg n}/\lg\lg n)$, one machine; $2^{\varepsilon n}$	superseded [25]
A5b	Hamiltonian cycle, one-sided error, $m \le Dn$, one machine; $2^{\varepsilon n}$	superseded [25]
A3	Vertex Cover; $C(n+2)^C 2^{\varepsilon k}$	known [17, 5]
A3′	Nonblocker; $C(n+2)^C 2^{\varepsilon k}$	known [15, 12]
N10	odd cycle transversal, deterministic, any density; $C2^{\delta n}$	known [17]

connected dominating set and k-leaf spanning trees on graphs with $m \le Dn$ and bounded degree, explicitly as a conjecture that is checked only numerically. He lists dominating set on dense graphs, and the exact counts of connected vertex covers and connected dominating sets, as not covered. His research notes also list the exact number of induced forests of each size for $m = O(n)$, and so the exact minimum feedback vertex set and the number of minimum ones, via a determinant identity checked numerically; it is not formalized. If it holds, it is deterministic and exact and so stronger than N9 and N9′. The two refutation papers list corollaries of their own: OpenAI [21, Cor. 1.3] counts independent sets, q-colorings for fixed q and perfect matchings (this covers N6), and Anthropic [1, Cor. 7] counts independent sets, cliques, vertex covers and dominating sets of each size, and k-colorings for fixed k, on graphs of any density.

Related work. The sparsification lemma [17] and the width–density duality of Calabro, Impagliazzo and Paturi [7] relate SAT variants with few variables; A1 is the latter duality combined with the claimed algorithm. Odd cycle transversal is in size-constrained SNP, so its decision version (N10) reduces to 3-SAT with $O(n)$ variables [17, Thm. 5]. For the maximum in A4, the equivalence of Max-r-SAT and Max-Cut with linearly many constraints with ETH is due to Jonsson, Lagerkvist, Schmidt and Uppman [18, Thm. 8]; Ganian, de Haan, Kanj and Szeider [15] show the same for Nonblocker and Dominating Set. For connectivity problems the direction we need was

open: Impagliazzo, Paturi and Zane list Hamiltonian circuit only as SNP-hard [17], and Ganian et al. ask whether Feedback Vertex Set reduces to 3-CNF-SAT in this way [15]; N9′ does not answer this, since it reduces sparse instances to sized counting, with randomness. Under ETH, connected dominating set has no $2^{o(n)}$ algorithm even on intersection graphs of convex objects in the plane [11]. For connected G with $n \geq 3$ vertices, n minus the connected domination number is the maximum number of leaves of a spanning tree [4]. Dell, Husfeldt, Marx, Taslaman and Wahlén [13] prove #ETH lower bounds for the permanent and the Tutte polynomial in terms of the number of edges or nonzero entries; N3, N6 and N6′ show that, given the claimed refutation, these problems become subexponential in the same parameter. Cut & Count [10] is the source of the parity reductions, and the isolation lemma of Calabro, Impagliazzo, Kabanets and Paturi [6] is the source of the isolation step in Section 3. Cai and Juedes [5] show that $2^{o(k)}\,\mathrm{poly}(n)$ for Vertex Cover refutes ETH; A3 is the converse, given the refutation. Cygan et al. [9] show that, under ETH, deciding $G \to H$ needs $|V(H)|^{\Omega(|V(G)|)}$ time in general. Our development uses Lean 4 [20] and Mathlib [19].

Organization. Section 2 describes the interface to Anthropic's development. Section 3 treats connected solutions on graphs of any density, Section 4 counting problems and odd cycle transversal, Section 5 Hamiltonian cycles and Section 6 parameterized problems, each with algorithms, theorems and proof sketches. Section 7 proves the barrier for Cut & Count, Section 8 describes the formalization and the reference implementation, and Section 9 discusses limitations.

2 Preliminaries and the interface

A literal is a pair (i, b) and holds under $x \in \{0,1\}^N$ iff $x_i = b$. A CNF is a list of clauses; its width is the maximum clause length. We write $|x|$ for the number of true variables, $\#\psi$ for the number of satisfying assignments of ψ on N variables, and $\mathrm{cnt}_s(\psi)$ for the number of those with $|x| = s$. Logarithms lg are to base 2.

Machines and encodings. A machine is a TM2 machine of Mathlib with finite stack alphabets; its time is the number of steps until it halts with the stated output. Inputs are bit strings. A graph on n vertices is 1^n0 followed by its adjacency matrix, a relation on $[h]$ is its h^2 entries row by row, a number t in unary is 1^t0, and a CNF is 1^n0 followed by its clauses, each literal written in $\lfloor \lg 2n \rfloor + 1$ bits.

The counters. We use three theorems of Anthropic's development as black boxes.

Theorem 1 (SIZED, Anthropic)**.** *For every width w and $\delta > 0$ there are a machine SIZED_w, given as a stepped computation with polynomial-time steps, and a constant c, such that for all N, $s \leq N$ and every CNF ψ of width at most w on N variables, SIZED_w on $(1^N, s, \psi)$ returns $\mathrm{cnt}_s(\psi)$ within $c \cdot 2^{\delta N}$ steps of size at most $c(N + |\psi| + 2)^c$.*

Theorem 2 (COUNT, Anthropic's Main Theorem)**.** *One machine COUNT and a constant C: for every $w \geq 1$ and every CNF ψ of width w with $m \geq 2$ clauses, none repeated, in which $N \geq 4$ variables occur, given in a compact code of length L, COUNT outputs $\#\psi$ within*

$$2^{C \min(wm/\sqrt{\lg m},\ w^2 N \lg\lg N/\sqrt{\lg N})} + CL^C$$

steps.

Algorithm 1 Parity of connected solutions (N2). Lean: `Conn.connF`, `ConnTM.connM`.

Input: graph G on n vertices, $1 \le \ell \le n$, CNF φ of width r on the vertex bits

1: x_v: $v \in X$; $s_v = x_{n+v}$: the side of v; $\bar{s}_v = x_{2n+v}$
2: $\psi \leftarrow \varphi$, $\bar{s}_v = \neg s_v$, $x_v \vee \neg s_v$, and for each edge uv: $x_u \wedge x_v \to (s_u = s_v)$
3: $c \leftarrow \mathsf{SIZED}_{r+4}(3n,\ n+\ell,\ \psi)$ ▷ width $\le \max(r,4)$; $c = \sum_X 2^{\mathrm{cc}(G[X])}$
4: **return** $\lfloor c/2 \rfloor \bmod 2$

Theorem 3 (DOMCOUNT, Anthropic)**.** *For every $\delta > 0$ there are a stepped machine DOMCOUNT and c that output the number of dominating sets of size s of a graph on n vertices within $c \cdot 2^{\delta n}$ steps.*

The number of clauses enters Theorem 1 only through the polynomial size of a step. Most of our reductions exploit this: their formulas have up to n^2 clauses, or as many as the input has edges, but only linearly many variables.

Stepped computations and hosts. Anthropic's machines are *stepped*: a computation is cut into polynomial-time steps, and a host theorem turns a stepped computation with T steps of state size S into a TM2 machine with $q(T+1)(|x|+S+1)^q$ steps. A depth-first walk over the tree of a step function that sums a stepped computation over the leaves is again stepped (`walkSum`); its number of steps is bounded via the number of leaves. We add one host.

Lemma 4 (FPT host, `tm_fpt`)**.** *Let a stepped computation with polynomial-time steps run on inputs x of size $n(x)$ with parameter $p(x)$ in at most $c\,2^{\delta p(x)}$ steps, $\delta \ge 0$, where the input length plus the state size is at most $a(n+2)^e$. Then there are C and one TM2 machine with finite alphabets that produces the same output within $\lfloor C(n+2)^C 2^{\delta p(x)} \rfloor$ steps.*

The pattern. Every algorithm below has the form

$$x \xrightarrow{\text{front end, poly time}} (N, s, \psi) \xrightarrow{\mathsf{SIZED}\text{ or }\mathsf{COUNT}} \text{count} \xrightarrow{\text{post-step}} \text{answer},$$

or a tree walk over such leaves. The front end is a polynomial-time function, proved so in Lean (`PolyT`), and it is joined to the counter by `Stepped.mapIn` and `Stepped.mapOut`. The proof obligations are always the same: the formula has the claimed width and number of variables, and its (sized) count is the quantity in the statement.

3 Connected solutions and induced forests

Cut & Count [10] replaces connectivity by the parity of a count. A vertex set X has $2^{\mathrm{cc}(G[X])}$ maps $s : X \to \{0,1\}$ that are constant on the components of $G[X]$, and $2^{\mathrm{cc}(G[X])-1}$ is odd iff $G[X]$ is connected. The side bits add n variables, and the edges of G enter only as clauses. Theorem 1 is insensitive to the number of clauses, so the graph may be dense.

3.1 The parity of connected solutions (N2)

Theorem 5 (`conn_parity_tm`, `cvc_parity_tm`)**.** *For every width r, exponent D and $\delta > 0$, one machine, given a graph G on n vertices of any density, $1 \le \ell \le n$ and a CNF φ on the vertices with clauses of at most r literals and at most $(n+2)^D$ clauses, outputs the parity of the number of $X \subseteq V$ with $|X| = \ell$, $X \models \varphi$ and $G[X]$ connected, within $C2^{\delta n}$ steps.*

Algorithm 2 Connected solutions with one-sided error (N2′). Lean: `ConnI.ciF`, `ConnITM.preR`.

Input: G, ℓ, φ as in Algorithm 1; a seed ω with $3n+3$ checks on the n vertex bits

$\psi \leftarrow$ the formula of Algorithm 1 on $3n$ variables; $z = x_{3n}$, add $\neg z$

for each check $i < 3n+3$ with terms $(j_0, c_0), \dots, (j_{k-1}, c_{k-1})$ **do**

  add $a_{i,l} = a_{i,l-1} \oplus (c_l \wedge x_{j_l})$ for $l < k$, where $a_{i,l} = x_{3n+1+ik+l}$ and $a_{i,-1} = z$

add $a_{i,k-1} =$ (right-hand side of check i) for the checks in use, and $x_j = \mathrm{bit}_j$ for the fixed coordinates

for every chain bit a, a complement bit $\bar{a} = \neg a$ ▷ keeps the target size fixed

$c \leftarrow \mathsf{SIZED}_{r+4}(3n+1+2(3n+3)k,\ n+\ell+(3n+3)k,\ \psi)$

return $\lfloor c/2 \rfloor \bmod 2$

Proof sketch. Each pair $(s_v, \bar{s}_v)$ has exactly one true bit, so the solutions of size $n+\ell$ are the pairs (X, s) with $|X| = \ell$, $X \models \varphi$ and s constant on the components of $G[X]$ and 0 outside X (`cnt_stepA`, `cnt_stepB`). There are $2^{\mathrm{cc}(G[X])}$ of them for each X, and $2^{\mathrm{cc}-1}$ is odd iff $G[X]$ is connected (`sig_parity`). The formula has $3n$ variables whatever the number of edges. □

With φ the unit clauses x_t for a terminal set T, the sets are the vertex sets of Steiner trees on ℓ vertices; with $\varphi = \{x_u \vee x_v : uv \in E\}$, the connected vertex covers of size ℓ (a separate machine on input G, ℓ).

3.2 Isolation for arbitrary sets

Theorem 6 (`isolation`; after [6])**.** *For every $\delta > 0$ there are k, t such that for all $M \geq M_0(\delta)$ and every nonempty $S \subseteq \{0,1\}^M$, at least a $2^{-\delta M}$ fraction of the seeds leaves exactly one point of S. A seed consists of $3M+3$ parity checks on k coordinates each (with right-hand sides), a number of them to use, and for every coordinate a value in $\{0, \dots, t-1\}$ and a bit; the coordinate is fixed to the bit when the value is 0.*

Proof sketch. One check on k coordinates fails to separate two points at distance d with probability $(1 + (1 - d/M)^k)/2$ (`two_mul_card_rv`, by flipping one coefficient). Call $x \in S$ semi-isolated if no point of S at distance $\geq \rho$ passes the same first m checks. For the least m with $2|S|q^m \leq 1$, half of the check arrays make a given x semi-isolated (`card_not_semi`); the semi-isolated points with the same check values lie in one Hamming ball (`card_fiber_le`). If the survivors lie within distance $< 2\rho$ of each other, fixing the coordinates where the two furthest differ leaves one point (`stepII`). The count `iso_count` is exact; `iso_real` takes $\rho = \lfloor M/a \rfloor + 1$ and $k = ab$, and the theorem picks t, b, a against δ. □

[6] state the lemma for solution sets of k-CNFs; its proof uses only $S \neq \emptyset$, and that is what we formalize. A check on k coordinates is a chain of k auxiliary variables $a_l = a_{l-1} \oplus (c_l \wedge x_{j_l})$ in clauses of width ≤ 3, and the chain bits are determined by the main variables. So isolating among the solutions of a CNF adds $O(Mk)$ variables, linear in M.

3.3 Steiner tree and connected vertex cover with one-sided error (N2′)

Theorem 7 (`conn_randomized`)**.** *For every r, D, $\delta > 0$ and $\varepsilon > 0$ there are k, t, n_0, C and one machine with the following property. For every graph G on $n \geq n_0$ vertices of any density, $1 \leq \ell \leq n$, every CNF φ as in Theorem 5 and every seed ω, the machine halts on $(G, \ell, \omega, \varphi)$ within $C2^{\varepsilon n}$ steps with a bit* $\mathrm{out}(\omega)$*. If no vertex set of size ℓ satisfies φ and induces a connected subgraph,* $\mathrm{out}(\omega) = 0$

Algorithm 3 Parity of connected dominating sets (N7). Lean: `CdsTree.rootN`, `CdsTM.cdsM`.

Input: graph G on n vertices, $1 \le \ell \le n$

open $\leftarrow \{N[v] : v \in V\}$; closed $\leftarrow$ the clauses of Algorithm 1 for $\varphi = \emptyset$, on $3n$ variables

$c \leftarrow 0$

for each leaf (open′, closed′) of the tree of the shortening step with width w **do**

    $c \leftarrow c + \mathsf{SIZED}_{w+4}(3n,\ n+\ell,\ \text{open}' \text{ as positive clauses} \cup \text{closed}')$

return $\lfloor c/2 \rfloor \bmod 2$

for every seed; if one does, $\mathrm{out}(\omega) = 1$ *for at least a* $2^{-\delta n}$ *fraction of the seeds. In particular this decides with one-sided error whether* G *has a Steiner tree on* ℓ *vertices and whether it has a connected vertex cover of size* ℓ*.*

Proof sketch. Algorithm 2. A chain bit and its complement hold exactly one true bit, so the solutions with $n+\ell+(3n+3)k$ true variables are those of Algorithm 1 with $n+\ell$ true variables whose vertex set passes the checks and the fixings (`stepC`–`stepE`), and half their number has the parity of the number of connected solutions of size ℓ that pass them (`ci_parity`). If there is no connected solution, this parity is 0 for every seed. If there is one, Theorem 6 applied to the set of connected solutions in $\{0,1\}^n$ leaves exactly one for a $2^{-\delta n}$ fraction of the seeds, and there the parity is 1 (`ci_decide`). The formula has $3n + 1 + 2(3n+3)k = O(n)$ variables, whatever the number of edges. □

Running the machine on R independent seeds and answering yes if some run outputs 1 never errs on a no-instance. On a yes-instance it errs on at most a fraction $e^{-R2^{-\delta n}}$ of the R-tuples of seeds, which is at most $e^{-2^{\delta n}}$ for $R \ge 2^{2\delta n}$ (`conn_amplified`).

3.4 Connected dominating set and Max Leaf (N7, N7′, N7″)

A connected dominating set is a set X with $G[X]$ connected such that every vertex is in X or has a neighbour in X. Domination is the CNF $\varphi_{\mathrm{dom}} = \{\bigvee_{u \in N[v]} x_u : v \in V\}$, but its clauses have unbounded width, so Theorem 5 does not apply. Anthropic's algorithm for dominating sets [1, Cor. 7] (Theorem 3) shortens these clauses with a search tree after Calabro, Impagliazzo and Paturi [7]: in a clause with more than w variables, let H be its first w variables; in the *light* child the clause is replaced by $\bigvee H$, in the *heavy* child the variables of H are false and are deleted from all open clauses. We run this tree with the clauses of domination open and the clauses of Cut & Count closed.

Theorem 8 (`cds_parity_tm`)**.** *For every* $\delta > 0$*, one machine, given a graph* G *on* n *vertices of any density and* $1 \le \ell \le n$*, outputs the parity of the number of connected dominating sets of* G *with* ℓ *vertices, within* $C2^{\delta n}$ *steps.*

Proof sketch. Every assignment satisfies exactly one child of a node, so the sized counts of the leaves add up to that of the root (`cntS_step`, `sum_leaves`), which is the formula of Algorithm 1 with $\varphi = \varphi_{\mathrm{dom}}$. A light step removes an open clause and a heavy step removes w of the variables that occur in open clauses. Anthropic's invariant bounds these variables by the number of variables of the formula; ours (`InvP`) bounds them by n while the formula has $3n$ variables and arbitrary closed clauses of width at most $w+4$. So a path has at most n light and n/w heavy steps, the tree has at most $\sum_{j \le 2n/w} \binom{2n}{j} \le 2^{\varepsilon n}$ leaves for w large enough, and every leaf has width at most $w+4$ (`shortensN`). With $\varepsilon = \delta/4$ and SIZED_{w+4} at $\delta/12$ on $3n$ variables, the walk takes $C2^{\delta n/2}$ steps; Theorem 5's parity argument (`conn_parity`) gives the output. □

Theorem 9 (`cds_randomized`). *For every $\delta > 0$ and $\varepsilon > 0$ there are k, t, n_0, C and one machine that, for every graph G on $n \geq n_0$ vertices of any density, $1 \leq \ell \leq n$ and every seed ω as in Theorem 7, halts on (G, ℓ, ω) within $C2^{\varepsilon n}$ steps with a bit $\mathrm{out}(\omega)$. If G has no connected dominating set of size ℓ, $\mathrm{out}(\omega) = 0$ for every seed; if it has one, $\mathrm{out}(\omega) = 1$ for at least a $2^{-\delta n}$ fraction of the seeds.*

Proof sketch. The closed clauses at the root are now those of Algorithm 2 for $\varphi = \emptyset$, on $3n{+}1{+}2(3n{+}3)k$ variables; the open ones are still the n clauses of domination on the vertex bits, so the tree and its bound are those of Theorem 8. The sum over the leaves is the count of Algorithm 2 for $\varphi = \varphi_{\mathrm{dom}}$, and `ci_decide` gives the one-sided error. Amplification is as for N2′ (`cds_amplified`). □

Theorem 10 (`maxleaf_randomized`, `mincds_randomized`). *For every $\delta > 0$ and $\varepsilon > 0$ there are k, t, n_0, C and one machine that, for every graph G on $n \geq n_0$ vertices of any density, every $1 \leq L < n$ and every seed ω, halts on $(G, n - L, \omega)$ within $C2^{\varepsilon n}$ steps with a bit $\mathrm{out}(\omega)$. If G has no spanning tree with at least L leaves, $\mathrm{out}(\omega) = 0$ for every seed; if it has one, $\mathrm{out}(\omega) = 1$ for at least a $2^{-\delta n}$ fraction of the seeds. On (G, ℓ, ω) with $1 \leq \ell \leq n$ a machine with the same bounds decides, in the same sense, whether G has a connected dominating set with at most ℓ vertices.*

Proof sketch. Lean states the two parts separately; both proofs take the machine of Theorem 9. Two facts about graphs, proved with Mathlib's walks and trees, translate the conditions. First, if X is a connected dominating set with $|X| < \ell \leq n$, some vertex outside X has a neighbour in X, and adding it keeps X connected and dominating; so a connected dominating set with at most ℓ vertices exists iff one with exactly ℓ does (`cds_le_iff`). Second, for $n \geq 3$ and $1 \leq L < n$, G has a spanning tree with at least L leaves iff it has a connected dominating set with $n - L$ vertices [4] (`leaf_iff`). In a spanning tree on $n \geq 3$ vertices the non-leaves induce a connected subgraph, because the inner vertices of a path have degree at least 2, and they dominate, because the neighbour of a leaf is not a leaf. Conversely, a spanning tree of $G[X]$ together with one edge from each vertex outside X to a neighbour in X is a spanning tree of G in which every vertex outside X is a leaf. A disconnected graph has neither, so no test of connectivity is needed. □

The proof of Theorem 8 uses only that the open clauses are positive, have no repeated variables and live on the vertex bits. With the invariant bounding the number of open clauses by Dn and their variables by n separately, as for A1, a path of the tree has at most Dn light and n/w heavy steps, and the tree of depth $(D+1)n$ has at most $\sum_{j \leq 2n/w} \binom{(D+2)n}{j} \leq 2^{\varepsilon n}$ leaves for a suitable w (`shortensP`).

Theorem 11 (`pos_parity_tm`, `pos_randomized`). *For every D and $\delta > 0$ there are C and one machine that, given a graph G on n vertices of any density, $1 \leq \ell \leq n$ and a CNF φ on the vertices with only positive literals, at most n literals per clause and at most Dn clauses of any width, outputs within $C2^{\delta n}$ steps the parity of the number of sets X with $|X| = \ell$ that meet every clause of φ and induce a connected subgraph. For every D, $\delta > 0$ and $\varepsilon > 0$ there are k, t, n_0, C and one machine that, for $n \geq n_0$ and every seed ω as in Theorem 7, halts on $(G, \ell, \omega, \varphi)$ within $C2^{\varepsilon n}$ steps with a bit $\mathrm{out}(\omega)$, which is 0 for every seed if no such X exists and 1 for at least a $2^{-\delta n}$ fraction of the seeds if one does.*

Proof sketch. The front end keeps the variables of each clause without repeats; a positive clause has the same models as the clause of its variables (`cntS_opens`). The rest is the proof of Theorems 8 and 9 with φ in place of φ_{dom}. □

With φ the groups, the sets X are the vertex sets of the group Steiner trees with ℓ vertices; with φ a set system, the connected hitting sets of size ℓ; with φ the open neighbourhoods, the connected

Algorithm 4 Parity of feedback vertex sets (N9). Lean: `Forest.fvsF`, `ForestTM.forM`.

Input: graph G on n vertices with edges $e_0, \dots, e_{m-1}$, $m \le Dn$; $s = n - k$

1: $\psi \leftarrow$ the formula of Algorithm 1 for $\varphi = \emptyset$, on $3n$ variables
2: **for** each edge $e_i = uv$ **do** add $b_i \to x_u$, $b_i \to x_v$, $\bar{b}_i = \neg b_i$, where $b_i = x_{3n+i}$, $\bar{b}_i = x_{3n+m+i}$
3: $c \leftarrow \mathsf{SIZED}_4(3n + 2m,\ n + s + m,\ \psi)$ $\quad \triangleright\ c = \sum_{|X|=s} S(G[X]) \cdot 2^{e(G[X])}$
4: **return** $\lfloor c/2^s \rfloor \bmod 2$

total dominating sets of size ℓ. In each case at most Dn sets are allowed. Theorem 5 allows negative literals, but only in clauses of bounded width.

For a positive φ, adding a neighbour to a connected set that meets φ gives another one, and in a connected graph some vertex outside a set $X \neq \emptyset, V$ has a neighbour in X. So a solution with at most $\ell \le n$ vertices grows to one with exactly ℓ (`conn_grow`, `min_iff`), and on connected graphs the machines of Theorems 7 and 11, run with ℓ, decide with one-sided error whether a solution with at most ℓ vertices exists (N8″, `min_conn_randomized`, `min_pos_randomized`; Lean states these as separate existence statements whose proofs use those machines): minimum Steiner tree (in vertices), minimum connected vertex cover and minimum Group Steiner tree.

3.5 Feedback vertex set on sparse graphs (N9, N9′)

A feedback vertex set is a set X such that $G - X$ is a forest. Its complement is an induced forest, so we count induced forests. Cut & Count for forests needs one more ingredient: edge bits.

Lemma 12 (`sides_edges`)**.** *Let H be a graph on a finite set W with e edges, and let S be the number of maps $W \to \{0,1\}$ that are constant on the edges of H. Then $S \cdot 2^e = 2^{|W|+j}$ for some $j \ge 0$, and $j = 0$ iff H is acyclic.*

Proof sketch. Induction on e; delete an edge uv. If u and v are not joined in $H - uv$, flipping the side of everything reachable from v shows that half the maps of $H - uv$ are maps of H (`card_sides_eq`), and H is acyclic iff $H - uv$ is. Otherwise every map of $H - uv$ is a map of H, the edge closes a cycle, and j grows by one. Mathlib's lemmas on adding an edge to an acyclic graph give both cases. □

Theorem 13 (`fvs_parity_tm`, `fvs_randomized`)**.** *For every D and $\delta > 0$ there are C and one machine that, given a graph G on n vertices with at most Dn edges and $n - k$ in unary, $k \le n$, outputs within $C2^{\delta n}$ steps the parity of the number of feedback vertex sets of G with k vertices. For every D, $\delta > 0$ and $\varepsilon > 0$ there are k_0, t, n_0, C and one machine that, for $n \ge n_0$, $k \le n$ and every seed ω as in Theorem 7, halts on $(G, n - k, \omega)$ within $C2^{\varepsilon n}$ steps with a bit $\mathrm{out}(\omega)$, which is 0 for every seed if G has no feedback vertex set with at most k vertices and 1 for at least a $2^{-\delta n}$ fraction of the seeds if it has one.*

Proof sketch. For a vertex set X the edge bits can be set freely on the edges inside X and must be 0 elsewhere, and each pair $(b_i, \bar{b}_i)$ has one true bit. So the solutions with $n + s + m$ true variables number $\sum_{|X|=s} S(G[X]) \cdot 2^{e(G[X])}$ (`fvs_count`), which by Lemma 12 is $\sum_{|X|=s} 2^{s+j_X}$, and $\lfloor c/2^s \rfloor$ is odd iff the number of induced forests of size s is odd (`fvs_parity`). The machine reads s again after the count to keep bit s. The complements of the feedback vertex sets of size k are the induced forests of size $n - k$ (`numFVS_eq`). Induced forests are closed under removing vertices, so a feedback vertex set with at most k vertices exists iff an induced forest with exactly $n - k$ vertices does (`fvs_iff`). The one-sided error is that of Theorem 7, whose counting steps (`stepC`–`stepE`) we prove again for any formula (`iso_count`, `fi_decide`). The formula has $3n + 2m \le (2D + 3)n$ variables, plus $1 + 2(3n + 3)k_0$ for the checks. □

Algorithm 5 #CNF-SAT of any width (A1). Lean: `CnfTM.cnfM`.

Input: CNF φ on n variables, width bound w

root: variables x_{2u} ("u true"), x_{2u+1} ("u false"); closed clauses $x_{2u} \vee x_{2u+1}$, $\neg x_{2u} \vee \neg x_{2u+1}$; open clauses: each clause of φ as a set of positive variables
$total \leftarrow 0$
for each node of the tree, depth first **do**
  if an open clause K has more than w variables; $H :=$ its first w **then**
    light child: remove K from the open clauses, add $\bigvee_{i \in H} x_i$ to the closed ones
    heavy child: add $\neg x_i$ for $i \in H$, delete H from every open clause
  else $total \leftarrow total + \mathsf{SIZED}_w(2n,\, n,\, \text{open} \cup \text{closed})$
return *total*

The edge bits make the number of variables grow with m, so unlike N2 this needs a sparse graph.

4 Counting

4.1 CNF of any width (A1, A1′)

Theorem 14 (`cnf_counts`)**.** *For every D and $\delta > 0$, one machine counts the satisfying assignments of every CNF with n variables and at most Dn clauses, of any width and with no literal repeated within a clause, within $C2^{\delta n}$ steps.*

Proof sketch. Algorithm 5. Since $|x| = n$ forces exactly one of x_{2u}, x_{2u+1}, the solutions of size n at the root are the assignments of φ. A node's count is the sum of its children's (either some variable of H is true, or all are false), and every leaf has width $\leq w$. The tree is Anthropic's clause-shortening tree from their Dominating Set corollary; we replace its invariant, which needs at most n clauses, by one for Dn clauses. The tree has at most $2^{(2/w)\lg(ew(D+2))n}$ leaves, which is $2^{\delta n/2}$ for w large. □

Theorem 15 (`cnf_grow`)**.** *One machine: for every $\delta > 0$ there is n_0 such that on every CNF with $n \geq n_0$ variables and at most $n \cdot 2^{(\lg n)^{1/4}/(\lg\lg n)^2}$ clauses of any width, with no literal repeated within a clause, it outputs the number of satisfying assignments within $2^{\delta n}$ steps.*

Proof sketch. Algorithm 5 with the width read off the node: for N variables, $\ell = \lfloor \lg N \rfloor$ and $L = \lfloor \lg \ell \rfloor$, take $w = 2^{\lfloor L/4 \rfloor - \lfloor \lg L \rfloor} + 1 \approx \ell^{1/4}/L$, and count each leaf with COUNT. Its exponent $Cw^2 N \lg\lg N/\sqrt{\lg N}$ is at most $32CN/L = o(n)$, since $w^2L^2 \leq 16\sqrt{\ell}$. For $D \leq 2^s + 1$ with $s = (\lg n)^{1/4}/(\lg\lg n)^2$, so that $s \leq 4w/(L-1)$, the coefficient $(2/w)\lg(ew(D+2))$ of the tree bound is at most $(8 + 6\sqrt{w})/w + 8/(L-1) \to 0$. □

4.2 Colorings and homomorphisms (A2, N5)

Let H be a relation on $[h] = \{0, \ldots, h-1\}$ (loops allowed, not necessarily symmetric) and $\hom(G, H)$ the number of maps $f : V(G) \to [h]$ with $H(f(u), f(v))$ for every edge uv. For $H = K_k$ these are the proper k-colorings.

Theorem 16 (`colorings_subexp`, `hom_subexp`, `hom_subexp_graph`)**.** *One machine, given h in unary, the matrix of H and an n-vertex graph G, outputs $\hom(G, H)$ within $2^{\varepsilon n}$ steps, for every $\varepsilon > 0$, $0 < \theta < 1/6$, $n \geq n_0(\theta, \varepsilon)$ and all $h \geq 2$ with $\lg h \leq (\lg n)^{1/6-\theta}$. For $H = K_k$ the output is the number of proper k-colorings (`numColorings_eq`). Two further machines of the same kind take*

Algorithm 6 Homomorphisms $G \to H$ (N5; A2 is $H = K_k$). Lean: `Hom.homB`, `colB`.

Input: graph G on n vertices, relation H on $[h]$

1: $b \leftarrow \lceil \lg h \rceil$; the bits $x_{vb}, \dots, x_{vb+b-1}$ spell the image of v
2: for each v and each code $h \le c < 2^b$: clause "the bits of v do not spell c"
3: for each edge uv (both orders) and each (a, c) with $\neg H(a, c)$: clause "not (u spells a and v spells c)"
4: tautologies $x_i \vee \neg x_i$ for $i < nb$; unit clauses $x_{nb}, \dots, x_{nb+3}$
5: **return** $\mathsf{COUNT}(nb + 4, \psi)$

a simple graph H and output the number of Mathlib homomorphisms $G \to_g H$, and take k in unary and output the number of proper k-colorings.

Proof sketch. The satisfying assignments of the formula of Algorithm 6 are exactly the homomorphisms (`cntN_homB`). It has $N = nb + 4$ variables and width $2b$, so the exponent of Theorem 2 is $C \cdot 4b^2 \cdot N \lg\lg N / \sqrt{\lg N} = O(n(\lg n)^{-2\theta})$ for $b \le 2(\lg n)^{1/6-\theta}$, because $b^3 \le 8(\lg n)^{1/2-3\theta}$ and $\lg\lg N \le (\lg N)^\theta / (\theta \ln 2)$. The at most $n^2h^2 + O(nh)$ clauses only enter the polynomial factor. □

Taking $h = 2$ and H an edge with a loop at one end counts the independent sets of G in Mathlib's sense, for graphs of any density (`numHom_hInd`, `indep_subexp`).

4.3 The Max-SAT spectrum (A4) and the Potts model (N3)

Theorem 17 (`maxsat_spectrum`). *For every width r, density D and $\delta > 0$, one machine outputs, given a CNF φ with $m \le Dn$ clauses of at most r literals and $t \le m$, the number of assignments that satisfy exactly t clauses, within $C2^{\delta n}$ steps.*

Proof sketch. Use the $2n$ variables of A1 and one variable $y_j = x_{2n+j}$ per clause K_j, with clauses $y_j \leftrightarrow K_j$ (width $r + 1$), the literal (u, b) read on x_{2u}. A solution with $n + t$ true variables is an assignment of φ together with the indicator of its t satisfied clauses (`cntS_specF`). Return $\mathsf{SIZED}_{r+2}(2n + m,\, n + t,\, \psi)$. □

For graphs with $O(n)$ edges, the Max-Cut spectrum and the density of states of the ± 1 Ising model follow from N3 below with $q = 2$ (`maxcut_spectrum`, `ising_spectrum`). For a graph G and q colors let N_j be the number of colorings $\sigma : V \to [q]$ with exactly j monochromatic edges, $k(A)$ the number of components of (V, A), and $T(G; x, y) = \sum_{A \subseteq E} (x-1)^{k(A)-k(E)} (y-1)^{|A|+k(A)-n}$.

Theorem 18 (`potts_spectrum`, `potts_fk`, `tutte_hyperbola`, `tutte_tm`). *For every $q \ge 1$, D and $\delta > 0$, one machine outputs N_j, given G with $m \le Dn$ edges and $j \le m$, within $C2^{\delta n}$ steps. In every commutative ring $\sum_j N_j (1+v)^j = \sum_{A \subseteq E} q^{k(A)} v^{|A|}$, and in every field, if $(x-1)(y-1) = q$ and $x, y \ne 1$, then $(x-1)^{k(E)} (y-1)^n T(G; x, y) = \sum_j N_j y^j$.*

Proof sketch. Variables x_{vq+c} ("v has color c") and $y_i = x_{nq+i}$ ("edge i is monochromatic"), $nq + m$ in all. Clauses: each vertex has at least one color (width q) and at most one (width 2); for each edge (a, b) and color c, $x_{aq+c} \wedge x_{bq+c} \to y_i$ and $y_i \wedge x_{aq+c} \to x_{bq+c}$. The solutions with $n + j$ true variables are the colorings with j monochromatic edges (`cnt_pottsF`); return $\mathsf{SIZED}_{q+3}(nq + m,\, n + j,\, \psi)$. The identities are the Fortuin–Kasteleyn expansion [14]: expand $\prod_e (1 + v[e \text{ mono}])$ over $A \subseteq E$; the colorings constant on the components of (V, A) are $q^{k(A)}$ many (`card_const`). □

Algorithm 7 The $\{-1,0,1\}$ permanent (N6′; N6 returns $\mathsf{SIZED}_2(m,n,\psi)$ after line 2). Lean: `Perm.permF`, `SPerm.chainF`, `SPermTM.spermM`.

Input: $A, S \in \{0,1\}^{n\times n}$: $M_{ij} = 0$ if $\neg A_{ij}$, -1 if $A_{ij} \wedge S_{ij}$, 1 otherwise

1: $e_0, \dots, e_{m-1} \leftarrow$ the entries with $A_{ij} = 1$, row by row; one variable x_t each; **if** $m < n$ **return** 0
2: $\psi \leftarrow \{\neg x_s \vee \neg x_t : s < t,\ e_s, e_t \text{ share a row or a column}\}$ ▷ width 2
3: $g_0, \dots, g_{r-1} \leftarrow$ the indices t with S at e_t
4: **for** $b \in \{0,1\}$ **do**
5: $\quad \psi_b \leftarrow \psi \cup \{\neg z\} \cup \{p_k = p_{k-1} \oplus x_{g_k},\ q_k = \neg p_k : k < r\} \cup \{p_{r-1} = b\}$
$\quad$ where $z = x_m$, $p_{-1} = z$, $p_k = x_{m+1+k}$, $q_k = x_{m+1+r+k}$ ▷ width 3
6: $\quad N_b \leftarrow \mathsf{SIZED}_3(m+1+2r,\ n+r,\ \psi_b)$
7: **return** $N_0 - N_1$

For $q = 2$, the colorings with j monochromatic edges are the vertex sets whose cut has $m - j$ edges (`pottsSpec_two`), and the spin configurations with $\sum_{uv\in E} s_u s_v = 2j - m$ (`ising_spectrum`). N_0 is the number of proper q-colorings in Mathlib's sense (`pottsSpec_zero`). It is the chromatic polynomial at q, and $N_0 = (-q)^{k(E)}(-1)^n T(G; 1-q, 0)$ (`chromatic_tutte`). At the point $(0, 1-q)$ the theorem gives $T(G; 0, 1-q)$ (`flow_tutte`), which by Tutte's theorem [24] is $(-1)^{m-n+k(E)}$ times the number of nowhere-zero $\mathbb{Z}_q$-flows. Under #ETH, [13] prove lower bounds of the form $2^{\Omega(n/\operatorname{polylog} n)}$ for such evaluations on sparse graphs.

4.4 The permanent (N6, N6′)

Theorem 19 (`perm_fpt`, `sperm_fpt`, `permanent_signed`). *For every $\varepsilon > 0$ there are C and one machine that, given a bit b, n in unary and $A, S \in \{0,1\}^{n\times n}$, outputs $N_b = \#\{\sigma \in S_n : \forall i\ A_{\sigma(i),i} = 1,\ |\{i : S_{\sigma(i),i} = 1\}| \equiv b \pmod 2\}$ within $C(n+2)^C 2^{\varepsilon m}$ steps, where m is the number of ones of A. The permanent of M (Mathlib's `Matrix.permanent` over $\mathbb{Z}$) equals $N_0 - N_1$. For $S = 0$ a separate machine outputs the permanent of the 0/1 matrix A directly, in the same time.*

Proof sketch. An assignment of ψ with exactly n true variables chooses n ones of A with no two in a row or a column, i.e. a permutation σ with $A_{\sigma(i),i} = 1$ for all i; the map is a bijection (`cntS_permF`), and these σ are the nonzero terms of the permanent of the 0/1 matrix (`permanent_eq`). In ψ_b the chain bits are determined by the choice (`chain_det`), p_k is the parity of the first $k+1$ entries -1 that are chosen (`chainVal_eq`), and every pair (p_k, q_k) has exactly one true bit. So the solutions of size $n + r$ are the permutations of parity b (`cnt_sperm`). The term of σ in the permanent of M is $(-1)^{j(\sigma)}$, where $j(\sigma)$ is the number of entries -1 on σ (`permanent_signed`). The formula has $m + 1 + 2r \le 3m + 1$ variables, so SIZED_3 at $\delta = \varepsilon/3$ takes $c\, 2^{\varepsilon/3} 2^{\varepsilon m}$ steps, and Lemma 4 gives the machine. □

Under #ETH the permanent of $\{-1,0,1\}$ matrices has no $\exp(o(m))$ algorithm, and that of 0/1 matrices none in $\exp(o(m/\log m))$ [13, Thm. 1.3]. Ryser's formula [23] takes $O(2^n n)$ time.

4.5 Odd cycle transversal (N10)

An odd cycle transversal is a set X of vertices such that $G - X$ is bipartite. The complement bits of N2 decide it with one sized count.

Theorem 20 (`oct_tm`). *For every $\delta > 0$ there are C and one machine that, given a graph G on n vertices (of any density) and $n - k$ in unary, $k \le n$, decides within $C 2^{\delta n}$ steps whether G has an odd cycle transversal with at most k vertices.*

Algorithm 8 Parity of the number of Hamiltonian cycles (A5a, A5a′). Lean: `Ham.hcF`, `HamS.hcS`.

Input: graph G on n vertices with edges $e_0, \dots, e_{m-1}$

1: x_i: edge e_i is chosen; $s_v = x_{m+v}$: the side of v
2: for each v: exactly two chosen edges at v
   A5a: $\neg x_a \vee \neg x_b \vee \neg x_c$ for distinct edges at v, and "at least two" (width $\deg v$)
   A5a′: a sequential counter along the edges at v, two bits per (edge, end), width 3
3: for each edge $e_i = ab$: $x_i \to (s_a = s_b)$; $\neg s_0$; $x_i \vee \neg x_i$ for every variable
4: **return** $\mathsf{COUNT}(N, \psi) \bmod 2$ ▷ $N = m + n$ (A5a), $N = 5m + n + 2$ (A5a′)

Proof sketch. The formula has, for each vertex v, a bit x_v (v is kept), a color s_v and $\bar{s}_v = \neg s_v$, with $s_v = 0$ if v is not kept, and for each edge uv the clauses $\neg x_u \vee \neg x_v \vee s_u \vee s_v$ and $\neg x_u \vee \neg x_v \vee \neg s_u \vee \neg s_v$. Its solutions with $n + s$ true variables are the pairs of a set Y of s vertices and a proper 2-coloring of $G[Y]$, so $\mathsf{SIZED}_4(3n, n + s, \cdot)$ is positive iff some s vertices induce a bipartite subgraph (`oct_pos`, with Mathlib's `IsBipartite`). Induced bipartite subgraphs are closed under removing vertices, so for $s = n - k$ this holds iff an odd cycle transversal with at most k vertices exists (`oct_iff`). The formula has $3n$ variables for any number of edges. □

The decision version is known (see the related work); N10 is its formalized, deterministic form, and running it for $k = 0, 1, \dots$ gives the minimum. The count weights each transversal by the 2-colorings of the rest, so it does not give the number of transversals.

5 Hamiltonian cycles

Let $\mathrm{hc}(G)$ be the number of connected spanning subgraphs of G with all degrees 2. In Section 8 we prove that it is the number of Hamiltonian cycles in Mathlib's sense. Tulsyan [25] computes $\mathrm{hc}(G)$ exactly and deterministically, which supersedes the results of this section; we obtained them independently and state them briefly. The encoding is that of Section 3 with edge bits in place of vertex bits.

Theorem 21 (`hc_parity_subexp`, `hc_parity_little_o`)**.**
For each Δ, one machine decides whether a graph with all degrees at most Δ has an odd number of Hamiltonian cycles within $2^{\varepsilon n}$ steps for $n \geq n_0(\varepsilon)$. One machine decides it for all graphs of any degree with at most $n\,g(n)$ edges, within $2^{\varepsilon n}$ steps for $n \geq n_0(g, \varepsilon)$, for every $\varepsilon > 0$ and every g with $g(n)\,r(n) \to 0$, where $r(n) = \lg\lg(n+4)/\sqrt{\lg(n+4)}$. That is, $m = o(n\sqrt{\lg n}/\lg\lg n)$ edges suffice. This includes $m \leq n(\lg n)^{1/2-\theta}$ (`hc_parity_sparse`) and every degree bound Δ (`hc_parity_bounded`).

Proof sketch. Algorithm 8 is Cut & Count [10]. A chosen 2-factor with c cycles has 2^{c-1} side maps with $s_0 = 0$, which is odd iff $c = 1$ (`parity_sides`: flip the sides on a cycle that avoids vertex 0), so $\#\psi \equiv \mathrm{hc}(G) \pmod 2$ (`hc_parity`). The sequential counter has the same solutions on the edge and side bits (`cntN_hcS`). For A5a′ the bound of Theorem 2 with width 3 and $N = 5m + n + 2$ variables is $c\,2^{9cN\,r(N)}$, with r as in the theorem. Since $m \leq n^2$, we have $N + 4 \leq (n+4)^3$ and hence $r(N) \leq 3r(n)$ (`rate_three`). So the exponent is at most $27c(5n\,g(n) + n + 2)\,r(n)$, which is below $\varepsilon n/2$ for large n because $g(n)r(n) \to 0$ and $r(n) \to 0$ (`bound0_le`). □

Theorem 22 (`ham_randomized`)**.** *There is one machine such that for every D, $\delta > 0$ and $\varepsilon > 0$ there are k, t, n_0 with the following property. For every graph G with $n \geq n_0$ vertices and at most Dn edges and every seed ω of checks on the m edge bits, the machine halts on G and ω within $2^{\varepsilon n}$*

Algorithm 9 Vertex Cover (A3). Lean: `Vc.stepV`, `Vc.leafV`, `VcTM.vcM`.

Input: graph G, budget k; a degree d chosen from ε

total $\leftarrow 0$
for each node (k', E') of the tree from $(k, E(G))$, depth first **do**
  if $k' > 0$ and an end v of an edge of E' has $\geq d$ neighbours $N(v)$ **then**
    children: $(k' - 1,\ E' \setminus \delta(v))$ and $(k' - |N(v)|,\ E' \setminus \delta(N[v]))$
  else
    $V' \leftarrow$ the ends of the edges of E'
    if $|V'| \leq k'd$ **then** $total \leftarrow total + \mathsf{SIZED}_2(|V'|, \min(k', |V'|), \{x_u \vee x_v : uv \in E'\})$
return $[total > 0]$

Algorithm 10 Nonblocker (A3′). Lean: `NbTM.preN`, `NbTM.nbM`.

Input: graph G on n vertices, $k \leq n$

$J \leftarrow$ the vertices with a neighbour; $n' \leftarrow |J|$
if $2k \leq n'$ **then return** yes ▷ Ore: $G[J]$ has a dominating set of size $\leq n'/2$
else if $n' < k$ **then return** no
else return $[\mathsf{DOMCOUNT}(G[J],\ n' - k) > 0]$ ▷ $n' < 2k$ vertices

steps with a bit $\mathrm{out}(\omega)$. *If G has no Hamiltonian cycle,* $\mathrm{out}(\omega) = 0$ *for every seed; if it has one,* $\mathrm{out}(\omega) = 1$ *for at least a* $2^{-\delta n}$ *fraction of the seeds.*

Proof sketch. Add the checks and fixings of ω on the edge bits to the formula of A5a′, as in Algorithm 2 but without complement bits, since COUNT does not fix the number of true variables. The count is $\equiv$ the number of Hamiltonian cycles that pass the checks and fixings (`iso_parity`), and Theorem 6 with $\delta/(D+1)$ in place of δ, applied to the set of Hamiltonian cycles in $\{0,1\}^m$, leaves exactly one for a $2^{-\delta n}$ fraction of the seeds, as $m \leq Dn$ (`iso_decide`). The machine decodes k, the checks and the fixings from the input (`decode_seedCode`), so one machine serves every δ. The formula has $5m + n + 2 + (3m+3)k = O(n)$ variables. Repetition over R seeds amplifies as in Section 3 (`ham_amplified`). □

6 Parameterized by the solution size

Theorem 23 (`vc_fpt`)**.** *For every $\varepsilon > 0$, one machine, given a graph G on n vertices (of any density) and $k \leq n$, decides whether G has a vertex cover of at most k vertices within $C(n+2)^C 2^{\varepsilon k}$ steps.*

Proof sketch. Algorithm 9; no kernel is used. A node has a cover within budget iff one child has (`good_step`), since a cover misses v only if it contains $N(v)$. Along a path the budget drops by 1 or by at least d, so by Anthropic's tree bound there are at most $\sum_{j \leq (k+d)/d} \binom{k+d}{j} \leq 2^{\varepsilon'(k+d)}$ leaves (`leaves_le`). At a leaf all degrees are below d, so a cover of size $\leq k'$ leaves at most $k'd$ vertices on the edges (`card_V_le`), and SIZED_2 decides the leaf in $2^{\delta kd}$ steps (`cnt_leafV_pos`); $\delta = \varepsilon/(4d)$ makes this $2^{\varepsilon k/4}$. Lemma 4 turns the walk into one machine. □

The best known algorithm for Vertex Cover runs in $O(1.2738^k + kn)$ time [8]. The Nonblocker problem asks for k vertices whose complement dominates, i.e. for a dominating set of at most $n - k$ vertices.

Theorem 24 (`nb_fpt`). *For every $\varepsilon > 0$, one machine, given G on n vertices and $k \le n$, decides whether G has a dominating set of at most $n - k$ vertices within $C(n+2)^C 2^{\varepsilon k}$ steps.*

Proof sketch. Isolated vertices lie in every dominating set, so the question is the same for $G[J]$ and the bound $n' - k$ (`nb_iff`). $G[J]$ has no isolated vertex, so by Ore's theorem [22] it has a dominating set of at most $n'/2$ vertices: the complement of a minimum dominating set dominates (`ore`). Supersets of dominating sets dominate, so in the remaining case a set of size exactly $n' - k$ exists iff one of size at most $n'-k$ does (`nb_answer`). DOMCOUNT at $\delta = \varepsilon/2$ runs in $c\,2^{(\varepsilon/2)n'} \le c\,2^{\varepsilon k}$ steps. □

This is the linear kernel of [12] in its simplest form, with the claimed refutation as the exponential part.

7 A barrier for Cut & Count

Our connectivity results give parities (N2, N7, N8, N9) or one-sided error (N2′, N7′, N7″, N8′, N8″, N9′), never a deterministic decision. This section shows that the colorings behind Cut & Count cannot do better over the rationals unless they use about ℓ colors.

Fix q colors with weights $x_1, \dots, x_q$ in a commutative $\mathbb{Q}$-algebra, the same for every vertex. A coloring $s : X \to \{1, \dots, q\}$ that is constant on the edges of $G[X]$ assigns one color to each component, so

$$\sum_s \prod_{v \in X} x_{s(v)} = \prod_C p_{|C|}(x), \qquad p_k = x_1^k + \cdots + x_q^k,$$

where C ranges over the components of $G[X]$. If the color of a root $r \in X$ is fixed to a, the sum is $x_a^{|C_r|} \prod_{C \neq C_r} p_{|C|}(x)$, where C_r is the component of r. The side bits of N2 are the case $q = 2$ with a root, and with the sized counter the weights become polynomials in a size variable. A set of size ℓ thus contributes a product indexed by its component sizes, and a connected one contributes p_ℓ, or x_a^ℓ with a root.

Proposition 25 (`barrier`, `barrier_rooted`). *Let $q < \ell$. Then p_ℓ is a rational linear combination of products $p_{\mu_1} \cdots p_{\mu_r}$ with $\mu_1 + \cdots + \mu_r = \ell$, all $\mu_i \ge 1$ and $r \ge 2$, and x_a^ℓ is a rational linear combination of products $x_a^c\, p_{\mu_1} \cdots p_{\mu_r}$ with $c \ge 1$, $r \ge 1$ and $c + \mu_1 + \cdots + \mu_r = \ell$. Hence a $\mathbb{Q}$-linear map that vanishes on the contribution of every disconnected set of size ℓ also vanishes on the contribution of a connected one, with or without a root.*

Proof. Each x_a is a root of $\prod_b (T - x_b) = \sum_{j=0}^q (-1)^j e_j T^{q-j}$, so $x_a^\ell = -\sum_{j=1}^q (-1)^j e_j x_a^{\ell-j}$ for $\ell > q$ (`pow_eq`), and summing over a gives Newton's identity $p_\ell = -\sum_{j=1}^q (-1)^j e_j p_{\ell-j}$ (`psum_eq`). By Newton's identities $k e_k = \sum_{i=1}^k (-1)^{i-1} e_{k-i} p_i$ (Mathlib's `mul_esymm_eq_sum`), so by induction e_j is a rational combination of products of power sums of total degree j, with at least one factor if $j \ge 1$ (`esymm_mem`). Since $j \le q < \ell$, the remaining factor $x_a^{\ell-j}$ or $p_{\ell-j}$ has positive degree (`pow_mem`, `psum_mem`). □

On a graph with ℓ vertices the count over $X = V$ is $p_{\mu_1} \cdots p_{\mu_r}$ for the disjoint union of cliques $K_{\mu_1} + \cdots + K_{\mu_r}$ and p_ℓ for K_ℓ. So a fixed rational combination of such counts, with any weights and any size refinement, that is 0 on every graph without a connected set of size ℓ is also 0 on K_ℓ: it neither decides nor counts the connected sets of size ℓ. Reduction modulo 2 is not $\mathbb{Q}$-linear, and this is how Cut & Count obtains parities with $q = 2$. The bound is tight in q: with $x = (1, \zeta, \dots, \zeta^{\ell-1})$ for a primitive ℓ-th root of unity ζ, $p_k = 0$ for $\ell \nmid k$ and $p_\ell = \ell$, so the count is ℓ times the number

of connected sets. But ℓ colors cost $\lceil \lg \ell \rceil$ bits per vertex, and the counters give $2^{o(n)}$ only for $n \cdot o(\sqrt{\lg n}/\lg\lg n)$ variables, that is, for $\ell = 2^{o(\sqrt{\lg n}/\lg\lg n)}$, where the $n^\ell = 2^{o(n)}$ sets of size ℓ can be listed directly.

The proposition covers weights that are the same for all vertices, apart from the root, and combinations that are fixed in advance. It says nothing about vertex-dependent weights, about restrictions of X that change from term to term, as in isolation, where each seed restricts X differently (this is how N2′ decides with one-sided error), or about other encodings, such as Tulsyan's matrix-tree identity on sparse graphs [25]. It also shows why N9 needs a bit per edge: a count of this form depends only on the component sizes of $G[X]$, so it cannot tell an induced path from an induced cycle. N9 therefore adds a bit per edge, which limits it to $O(n)$ edges.

8 Formalization and reference implementation

The Lean development. The development consists of 50 Lean files with about 15,800 lines on top of Anthropic's development (pinned to a fixed revision) and Mathlib. A check file prints the axioms of 91 theorems, the main theorems and the lemmas cited in the proof sketches; each depends on `propext`, `Classical.choice` and `Quot.sound` only, and there is no `sorry`. Every main theorem quantifies over the machine existentially and states its finite alphabets, its output (`Wrap.Outputs`) and its time bound explicitly. The polynomial-time front ends are verified with Anthropic's `PolyT` combinators and `fun_prop`. Most of the proof effort goes into two kinds of lemmas: shape bounds (width, number of variables and clauses of the formula, as functions of the input) and count identities (the sized count equals the intended quantity), which are proved as explicit bijections of finite sets.

Checks against Mathlib. Lean cannot check that a definition means what is intended, so where Mathlib has an independent definition we prove agreement. The number of Hamiltonian cycles $\mathrm{hc}(G)$ is positive iff G satisfies Mathlib's `SimpleGraph.IsHamiltonian` (for $n \neq 1$), and it equals the number of edge sets of walks with Mathlib's `IsHamiltonianCycle` (`numHamCycles_pos_iff`, `numHamCycles_eq`); the proof uses Mathlib's result that a connected graph with all degrees 2 is a single cycle. The homomorphism count equals the number of Mathlib homomorphisms $G \to_g H$ for simple H (`numHom_simple`), both permanents are stated with Mathlib's `Matrix.permanent`, the spanning trees of N7″ are subgraphs with Mathlib's `IsTree`, the forests of N9 are induced subgraphs with Mathlib's `IsAcyclic`, and N10 uses Mathlib's `IsBipartite`. The remaining definitions (the spectrum, the Potts counts, the Tutte polynomial, the seeds and the input codes) must be read by hand, as must Anthropic's model of computation and statement files.

What is not formalized. The numbered theorems, Lemma 4, the three counters of Section 2, the special cases (independent sets, the Max-Cut spectrum, the Ising model, the chromatic polynomial and $T(G; 0, 1-q)$) and the amplification of the one-sided error are Lean theorems; the proof sketches summarize the Lean proofs. Not formalized are the results we cite from the literature (lower bounds, earlier algorithms, Tutte's identity between $T(G; 0, 1-q)$ and nowhere-zero flows), the identity between colorings and power sums in Section 7, which is tested in Python, and the novelty labels of Table 1.

The reference implementation. A Python package of about 2,400 lines mirrors every front end and post-step clause by clause, with the same variable indices and from the same input words as the Lean machines. Anthropic's counters are replaced by an exact DPLL counter with component

caching that returns the weight enumerator of the solutions, so the implementation runs in exponential time; its purpose is to make the reductions concrete and testable. The tests compare every algorithm with a direct enumeration of the quantity in the Lean statement (colorings, homomorphisms, assignments, covers, dominating sets, permutations, Hamiltonian cycles, connected vertex sets, connected dominating sets, spanning trees, induced forests, bipartite induced subgraphs), check the shape bounds that the Lean proofs use, and measure the one-sided error of A5b, N2′, N7′, N7″, N8′ and N9′ over random seeds. A further test checks that the Python front ends are the Lean ones: a small Lean program evaluates the Lean definitions of all 23 front ends on 130 random input words, and the formulas agree clause for clause.

Availability. The Lean development, the Python package and the instructions to check both are at `https://github.com/ssenge/eth-corollaries`.

9 Limitations and conclusion

What the theorems depend on. Our results are Lean theorems; they do not assume an unproved hypothesis. They are exactly as sound as Lean's kernel and Anthropic's formalization, and they say what is intended only if Anthropic's statement files and the model of computation do. At the time of writing nobody has reviewed these statements independently, and the claimed refutations themselves are under review. The algorithms are galactic: no gain over 2^n is expected below $n \approx 10^{10}$.

What the results show. The variable count is the currency. $O(n)$ variables give $2^{\varepsilon n}$ for every ε; $n \cdot r(n)$ variables with $r = o(\sqrt{\lg n}/\lg\lg n)$ still give $2^{o(n)}$; for the uniform bound the clause width must stay below about $(\lg n)^{1/4}$. A sized counter makes the number of clauses irrelevant, which is why dense graphs are possible for N2, N2′, N7, N7′, N7″, N8, N8′, N8″, N10, A3 and A3′, and complement bits make it compatible with auxiliary variables.

Open problems. Colorings and homomorphisms into targets with $\lg h \geq (\lg n)^{1/6}$; #CNF-SAT of unbounded width with more clauses than in A1′; a deterministic version of N2′, N7′ and N7″, that is, deterministic Steiner tree, connected vertex cover, connected dominating set and Max Leaf Spanning Tree on dense graphs, which by Proposition 25 needs more than Cut & Count with few colors; the exact number of connected vertex sets of size ℓ on dense graphs, of which N2 gives the parity; Hamiltonian cycle on graphs of average degree $2^{\Omega(\sqrt{\lg n})}$, beyond Tulsyan's bound [25]; positive CNFs of unbounded width with superlinearly many clauses, such as Group Steiner tree with $\omega(n)$ groups; feedback vertex set on graphs with $\omega(n)$ edges, and deterministically; the number of odd cycle transversals of each size; the Tutte polynomial off the integer hyperbolas; permanents with larger integer entries; the matching polynomial of general graphs; and problems for which the refutations remove a lower bound but no reduction with few variables is known, such as dense Max-Cut, CNF-SAT with poly(n) clauses, planar problems below $2^{O(\sqrt{n})}$ and Subset Sum.

Use of AI

Claude Opus 5.5 (Anthropic) was used throughout this research: to explore approaches, to develop and check proofs, to write the Lean formalization and the reference implementation, and to draft the text. All theorems were checked by Lean.

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